1 Introduction

1.1 Definetly include

- Define masting
- Introduce different hypotheses related to masting
- Introduce that different hypotheses act on different stages of the reproductive process (e.g., flowering, pollination, seed maturation, dispersal)
- Existing studies regarding functional traits and masting found:
  - High CV is common in species with high stem tissue density
  - Short-lived leaves with low LMA is associate with low CV
  - Large seed is associate with high CV
  - floral transition is associate with year-to-year variability
  - wind pollination is associate with hight CV
- Existing studies lack emphasis on the role of reproductive traits as mechanistic links between masting and hypotheses.
- Especially on relationship between seed mass and masting focus on an ecological constraint perspective, where seed mass reflects the cost of reproduction. However, we try to capture the evolutionary mechanisms underlying masting benefits. For masting to be adaptive under the predator satiation hypothesis, seeds must be consumed by animals. Therefore, fruit size might be more relevant than seed mass alone.
- Our study propose a framework connecting:
  - reproductive traits
  - stages of reproduction
  - specific masting hypotheses
- This framework can clarify which traits are under selection in masting species

2 Discussion

2.1 Definetly include

- Seed mass and wind pollination are reproductive traits significantly associated with masting patterns. This might suggest that reproductive allocation plays a key role in shaping masting behavior, supporting predation satiation hypothesis and pollination coupling hypothesis.

- However, the fact that we didn't find relationships with other traits indicates that trait-based explanations alone are not sufficient to fully explain masting.

- **Gymnosperm are more likely to be masting species**, (is there a possibility that masting might be an ancestral strategy that becomes lost in some angiosperm lineages later?)

- We found moderate phylogenetic signal for masting in both gymnosperms and angiosperms, which align with other people's finding (e.g.,: Pearse et al., 2020, found a modest phylogenetic signal in CVp), then link to next point.

- **Masting is strongly related to environment as well**, some species might be exhibit masting or not depending on the environment they are in, broad trait-based analysis do not capture environmental drivers, and that's the reason why we can't use any masting metrics capturing other characteristics of masting.

- Different hypotheses may act at different stages (for flower masting and seed masting species), playing the selection on different species, need better data on flowering traits and finer resolution on masting definition.

- Existence of clear counterexamples especially within the clade. Focusing on within-clade variation (oaks) may be more informative. Need more detailed monitoring on masting pattern and reproductive traits for more species in oak family, potentially with data for same species in different environment.